Gabriel **Pedde Ungureanu** | Physics PhD | Quantitative Researcher

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Profile

PhD physicist (6 papers, ~60 citations) with a mathematics degree (110/110 *cum laude*): probability, stochastic modelling, numerical PDEs. I turn mathematical structure into fast, validated code — Gaussian-process inference from first principles, nonlinear PDE solvers, reinforcement learning at EuroHPC scale.

Research & Engineering Experience

SISSA **Trieste, Italy**
Doctoral Researcher in Theoretical Physics *2022–2026*

- Exact, non-perturbative results in quantum field theory via large-scale symbolic computation — generating functions (Hilbert series), combinatorial dualization algorithms — on SISSA's Ulysses cluster; **6 published papers, ~60 citations**.
- Built an automated **"AI co-scientist"** research pipeline: LLM hypothesis generation refereed by a **deterministic symbolic/numerical evaluator** (the model never grades itself); recovers exact General Relativity solutions to **machine precision** through self-improving evolutionary search.

Education

SISSA / ICTP Joint Programme **Trieste, Italy**
Master in High Performance Computing and AI (MHPC) *2025–2026*
Concurrent with the last year of PhD. Numerical analysis for PDEs, Bayesian statistics, machine learning, deep learning, reinforcement learning, and distributed GPU computing.

Università Cattolica del Sacro Cuore **Brescia, Italy**
BSc Mathematics & MSc Physics *2017–2022*
Real & functional analysis, probability, PDEs; theoretical physics, QFT, statistical mechanics. MSc thesis in **quantum complexity theory**, published in Physical Review D (28 citations).

Awards & Publications

6 peer-reviewed papers in theoretical physics (~**60 citations**) — INSPIRE-HEP | Google Scholar

Best graduate of the year, MSc Physics — Università Cattolica del Sacro Cuore

110/110 *cum laude* — MSc Physics | 110/110 *cum laude* — BSc Mathematics

Istituto Toniolo merit fellowship — BSc, Università Cattolica del Sacro Cuore

Selected Projects — Quantitative Modelling

Bayesian time series: **Quasi-periodic Gaussian Process built from scratch** (no GP libraries) to extract stellar rotation periods from noisy NASA Kepler light curves: Cholesky-factorised **log-marginal-likelihood maximisation**, Lomb–Scargle initialisation, full posterior predictive mean and uncertainty bands | GitHub

Nonlinear PDEs: Solved a stiff **nonlinear parabolic PDE** (Wetterich renormalization-group flow) two independent ways: method-of-lines **finite differences** (adaptive RK45) and a **physics-informed neural network** (PyTorch Lightning); located its fixed point via a **shooting method**, recovering critical exponent $\nu = 0.649$ vs 0.630 (Monte Carlo). | GitHub

Reinforcement learning: Implemented **RLOO/PPO from scratch** to fine-tune Llama-3.2-1B on GSM8K math reasoning with a rule-based reward and KL regularisation; **+11 pp answer accuracy** (35 % → 46 %). Distributed on **Leonardo (CINECA)**. Stack: PyTorch, TRL, SLURM | GitHub

Performance engineering: CUDA kernel development through to cuBLAS; distributed dense linear algebra with **MPI + CUDA**. Lattice Boltzmann CFD solver: ~**120× GPU speedup** over CPU, 806 GB/s sustained bandwidth | GitHub

Technical Skills

Quant methods: Probability & stochastic processes, Bayesian inference, Gaussian processes, time series, Monte Carlo, numerical PDEs (FD/FEM), iterative solvers, optimisation

ML / AI: PyTorch, scikit-learn, reinforcement learning (REINFORCE, RLOO, PPO), CNNs, autoencoders, dimensionality reduction, clustering, LLM agents (LangGraph, RAG)

Programming: Python (NumPy, SciPy, pandas), C++, C, Fortran, Mathematica, Git, LaTeX

Scale / HPC: CUDA, cuBLAS, MPI, OpenMP, SLURM, PETSc/SLEPc, Docker/Apptainer; Leonardo (CINECA), Ulysses (SISSA)

Languages: Italian (native) | English (C1) | Romanian (A2)